

Study of $\text{CaSO}_4\text{:Tm,Li}$ Based Luminescent Composites Applied to TL/OSL Dosimetry in Mammography

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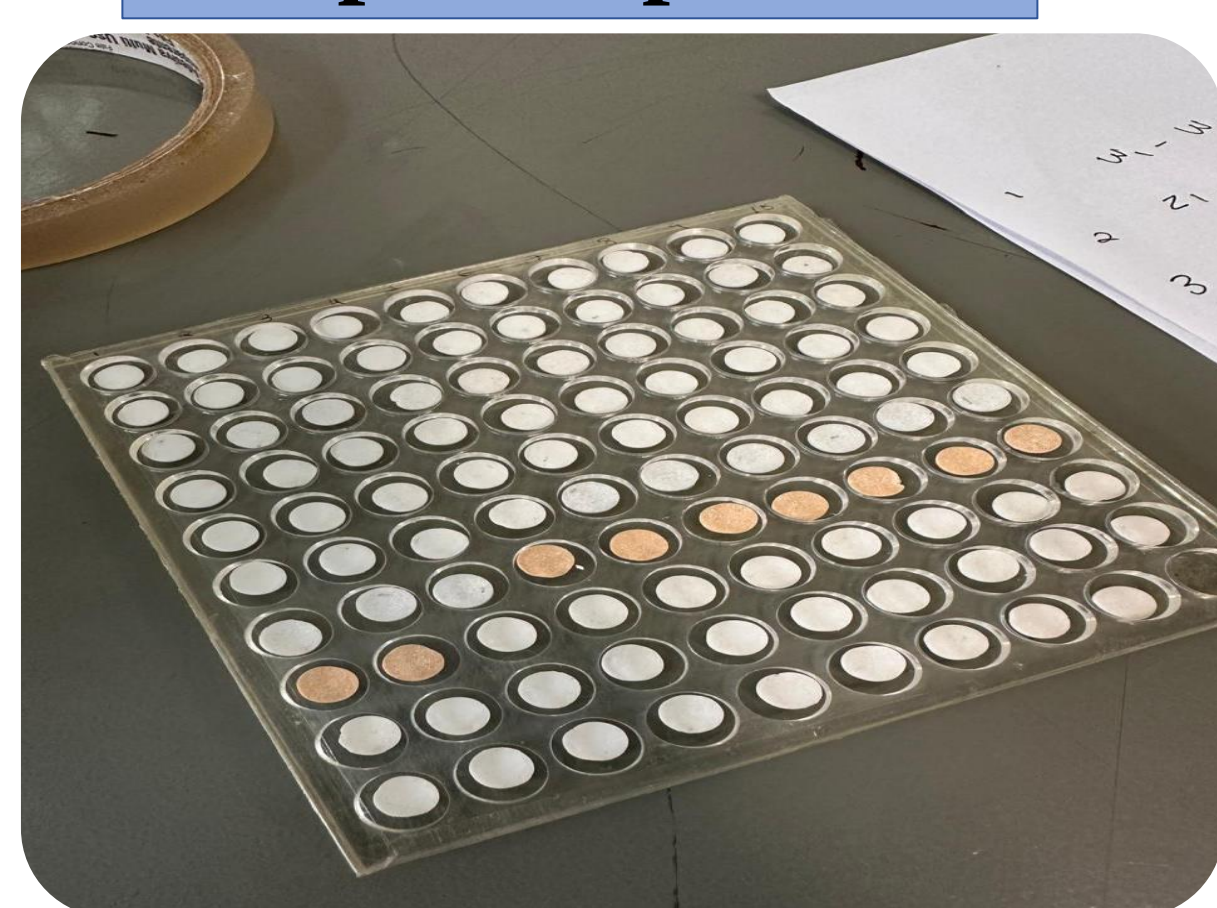
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INTRODUCTION

Mammography, through images obtained with X-rays, enables the early detection of nodules and other breast abnormalities, often before they can be detected by conventional methods such as self-examination. However, its implementation requires strict technical rigor, since failures in image quality may compromise diagnosis or delay the initiation of appropriate treatment [1]. With the advancement of digital mammography, especially with the introduction of tomosynthesis, an increase in radiation doses applied during examinations has been observed [2–3]. Given the potential risks associated with cumulative be identified by radiation exposure, it is essential to accurately monitor the absorbed dose in patients. This control must balance safety and effectiveness, ensuring that doses remain as low as reasonably achievable without compromising the image quality required to identify small variations in tissues of similar density. Like any examination that uses ionizing radiation, mammography involves a stochastic risk, albeit small, of cancer induction. Thus, the assessment of absorbed dose, particularly the mean glandular dose (MGD), becomes essential to ensure patient safety. In this context, dosimetry techniques such as thermoluminescence (TL) and optically stimulated luminescence (OSL) have proven effective in determining the entrance skin dose (ESD) and MGD in digital mammography units [4]. This study aims to evaluate the applicability of $\text{CaSO}_4\text{:Li}$, $\text{CaSO}_4\text{:Tm}^0$, $\text{CaSO}_4\text{:Tm}^{3+}$, $\text{CaSO}_4\text{:Tm}^0\text{:Li}$, and $\text{CaSO}_4\text{:Tm}^{3+}\text{:Li}$ phosphor composites as dosimeters using TL and OSL techniques in mammography beams. The response of these new materials will be compared with that of traditional $\text{CaSO}_4\text{:Dy}$ dosimeters, widely used in Brazil.

MATERIALS AND METHODS

Sample Preparation



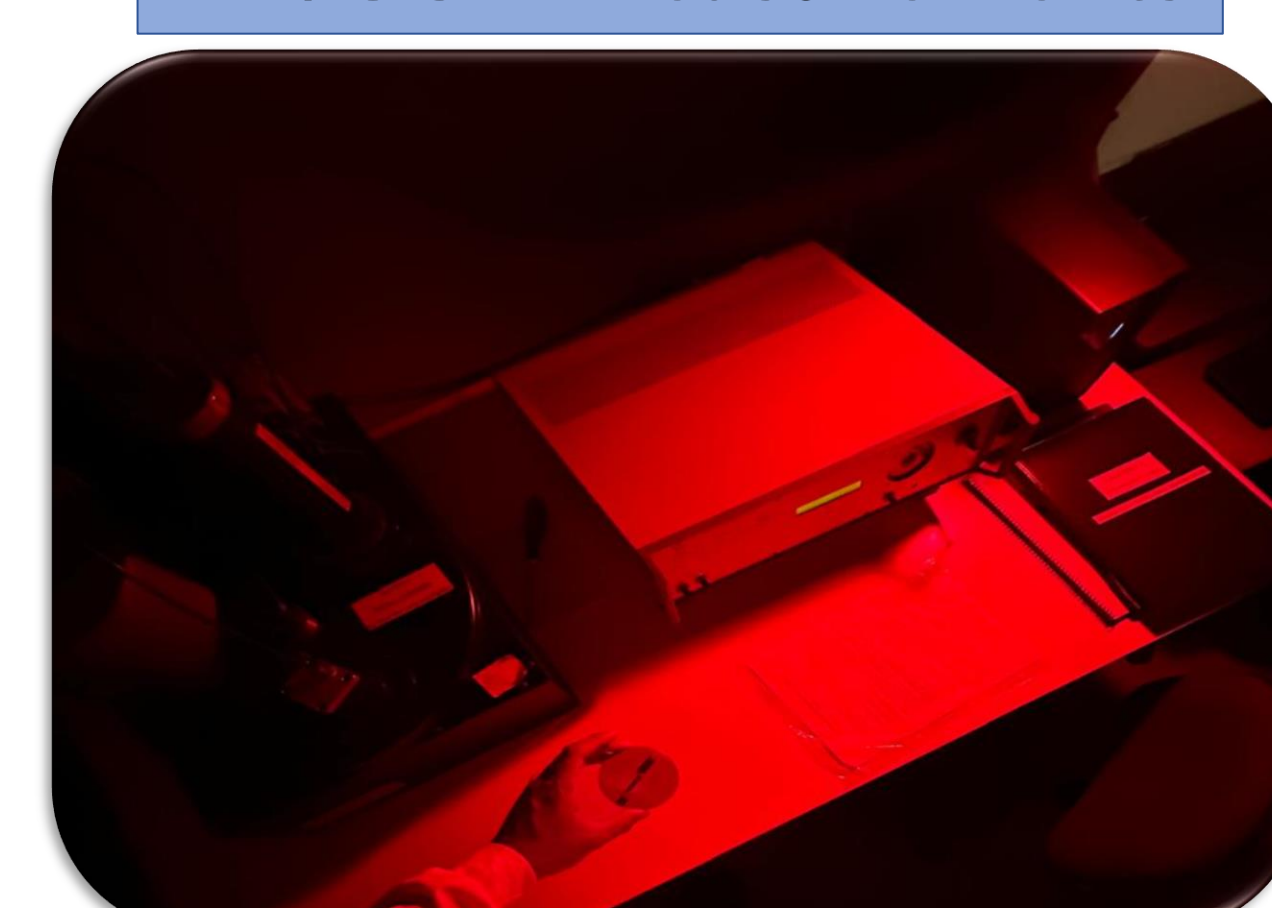
- Eight new groups prepared in collaboration between UFS and IPEN;
- Produced via the slow evaporation route (Silva et al. [5]);
- Samples in pellet form (CaSO_4 + Teflon, 1:1).

Irradiation



- Pantak-Seifert Isovolt HS X-ray equipment, 160 kV;
- Tungsten target + molybdenum filter + additional 2 mm Al filter (RQA-M2);
- Fixed distance: 1 m between source and samples;
- Beam conditions: 28 kV;
- Applied dose: 5 mGy;
- Conditions representative of digital mammography.

TL/OSL Measurements



- Risø TL/OSL-DA-20 system;
- TL analyses were carried out at a heating rate of 5 °C/s using:
- Schott BG-39 filter: transmission in the VIS region (320–480 nm);
- Hoya U-340 filter: transmission in the UV region (260–400 nm).
- OSL measurements were performed under continuous-wave stimulation (CW-OSL) with blue LEDs (470 nm), using an integration time of 40 s.

RESULTS

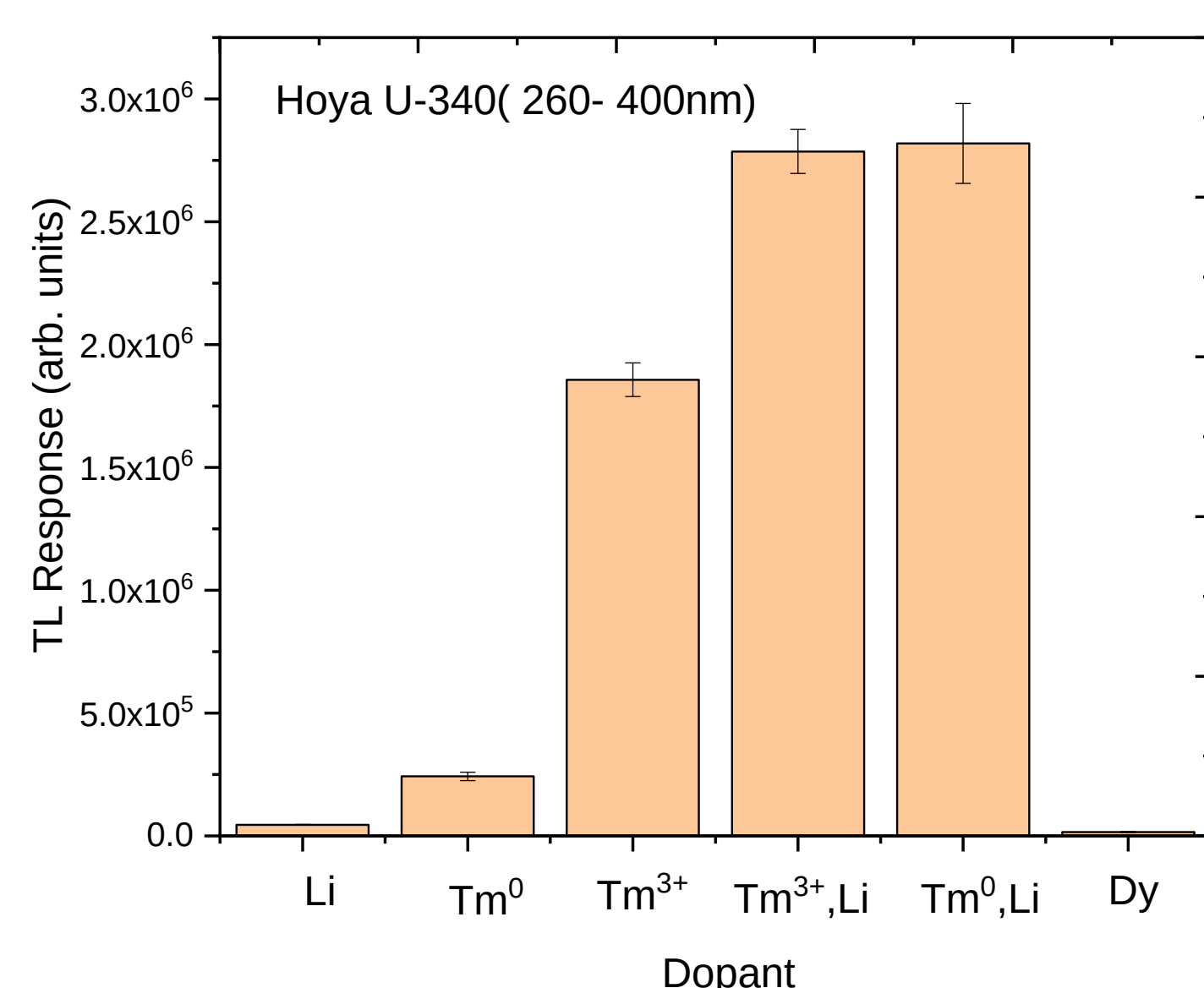


Figure 1. TL signal obtained by integrating the area under the emission curves of $\text{CaSO}_4\text{:Li}$, $\text{CaSO}_4\text{:Tm}^0$, $\text{CaSO}_4\text{:Tm}^{3+}$, $\text{CaSO}_4\text{:Tm}^0\text{:Li}$, and $\text{CaSO}_4\text{:Tm}^{3+}\text{:Li}$ samples irradiated with 5 mGy, using a Hoya U-340 filter.

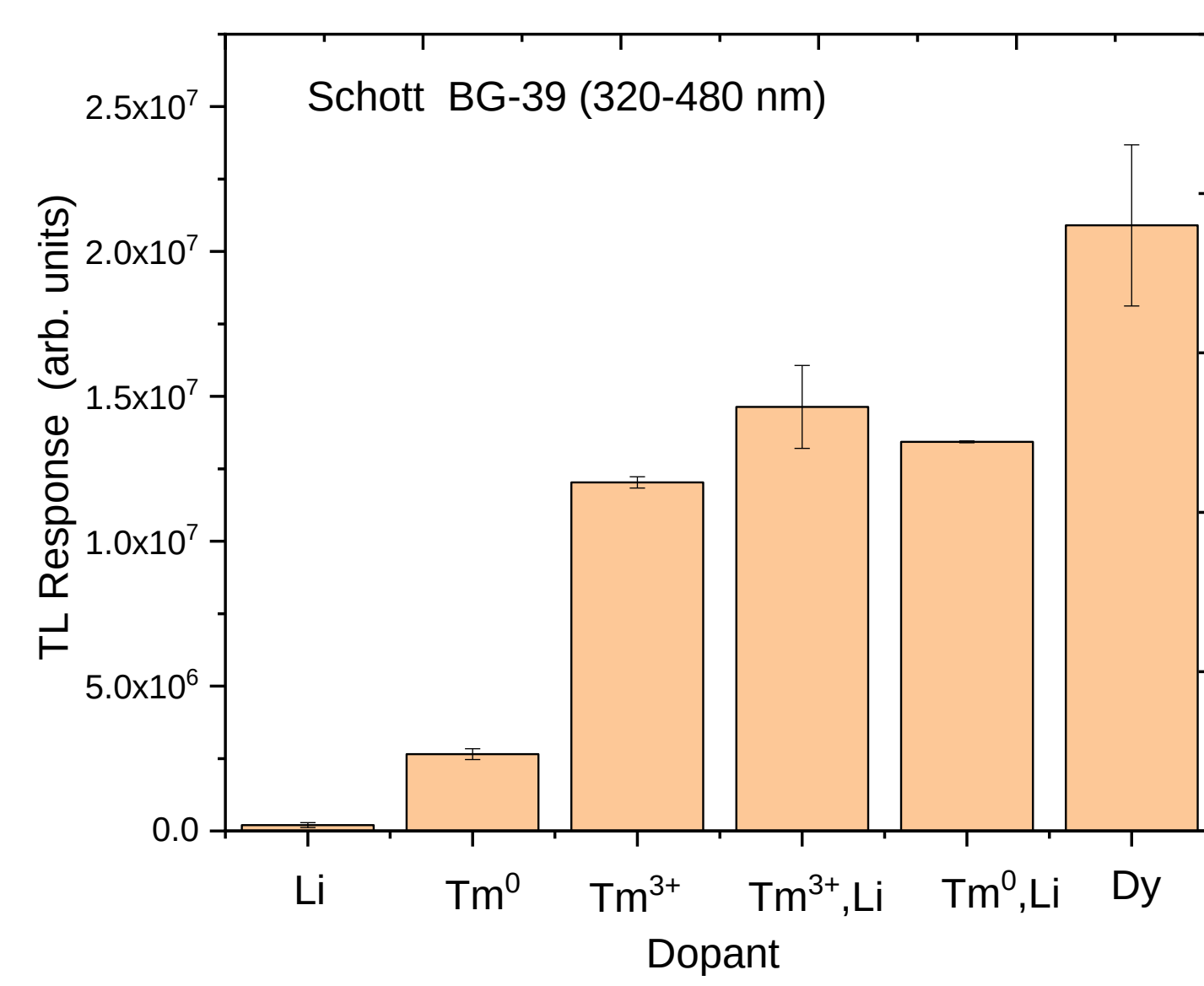


Figure 2. TL signal obtained by integrating the area under the emission curves of $\text{CaSO}_4\text{:Li}$, $\text{CaSO}_4\text{:Tm}^0$, $\text{CaSO}_4\text{:Tm}^{3+}$, $\text{CaSO}_4\text{:Tm}^0\text{:Li}$, and $\text{CaSO}_4\text{:Tm}^{3+}\text{:Li}$ samples irradiated with 5 mGy using a Schott BG-39 filter.

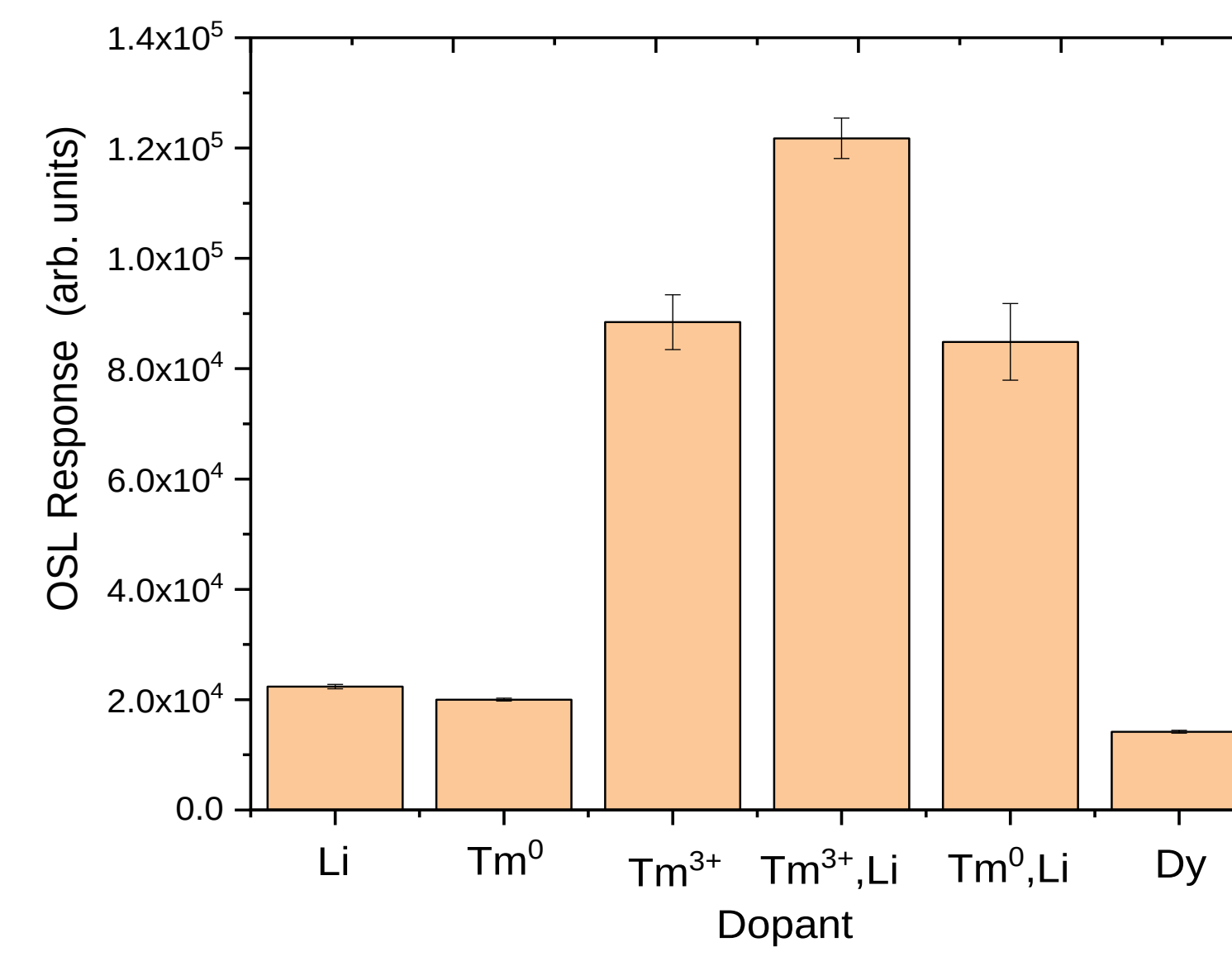


Figure 3. OSL signal obtained by integrating the area under the OSL decay curves of $\text{CaSO}_4\text{:Li}$, $\text{CaSO}_4\text{:Tm}^0$, $\text{CaSO}_4\text{:Tm}^{3+}$, $\text{CaSO}_4\text{:Tm}^0\text{:Li}$, and $\text{CaSO}_4\text{:Tm}^{3+}\text{:Li}$ samples irradiated with 5 mGy, using a Hoya U-340 filter.

Conclusion

Co-doped samples with lithium exhibited higher TL and OSL sensitivities compared to the monodoped ones. All tested materials demonstrated strong luminescent signals and low coefficients of variation, indicating good reproducibility and clinical feasibility for mammography dosimetry. Moreover, the intense OSL responses reinforce the potential of these composites for rapid optical readout systems. For typical mammography doses, the TL responses in the VIS region were comparable to those of standard $\text{CaSO}_4\text{:Dy}$ dosimeters, while the UV and OSL responses were superior, confirming the viability of these new materials as low-cost, high-performance alternatives for digital mammography dosimetry.

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